

GraspCLIP-D: Language-driven Grasp Detection via Dense Self-Supervised Features and Multi-Scale FiLM Fusion

Nguyễn Trần Minh Tâm
FPT AI Residency, Batch 7
nguyentranminhtam04@gmail.com

Abstract

*Language-driven grasp detection predicts an oriented grasp rectangle from an RGB image and a free-form prompt. The state of the art uses conditional diffusion, which is slow at inference. We revisit the task as discriminative dense prediction and propose **GraspCLIP-D**: frozen DINOv2 and CLIP encoders fused via multi-scale FiLM and a lightweight bottleneck cross-attention block, with only 10.3M trainable parameters. On Grasp-Anything++, it reaches $H = 0.60$ (Base = 0.74, New = 0.51) under the base-to-new protocol, surpassing the diffusion-based LGD (0.47) and consistency-based LLGD (0.46). Ablations show that dense self-supervised features and the bottleneck cross-attention are the two design choices that contribute most, while multi-scale FiLM helps only when paired with the global cross-attention mixer. Code: https://github.com/taamnguyen04/Alo_Grasp.*

1. Introduction

Robotic grasp detection has reached strong accuracy on object-level benchmarks [1, 2]. *Language-driven* grasp detection [8] additionally accepts a prompt specifying *which* object or part to grasp, the form most useful for human-facing applications.

The state of the art, LGD [8], uses conditional diffusion. While effective, diffusion requires up to 1000 denoising steps [3] — impractical for real-time robotics.

We propose a simpler discriminative formulation. First, instead of CLIP features — weak for dense tasks [9] — we use **DINOv2** [4] patch features. Second, we condition the decoder on language via multi-scale **FiLM** [5] and a bottleneck cross-attention. Our **GraspCLIP-D** needs only ~ 10 M parameters.

Contributions. (i) A discriminative architecture with DINOv2 dense features, multi-scale FiLM, and bottleneck cross-attention (10.3M trainable params). (ii) Evidence that dense DINOv2 features outperform CLIP, and that bottle-

neck cross-attention yields the largest gain. (iii) A fast single-pass alternative to diffusion pipelines.

2. Related Work

Grasp detection. GG-CNN [2] and GR-ConvNet [1] perform pixel-wise prediction of quality, angle, and width maps — the formulation we adopt.

Language-driven grasping. Vuong et al. [7] introduced Grasp-Anything. LGD [8] added part-level prompts via diffusion; LLGD [3] uses a consistency model. Recent discriminative methods explore hierarchical language-vision fusion. Our contribution is orthogonal: we show that pairing frozen DINOv2 features with a bottleneck cross-attention is sufficient, trading complexity for speed. Our method predicts grasps in a single pass.

Vision-language fusion. CLIP [6] aligns globally, but is weak for dense prediction [9]. DINOv2 [4] features transfer strongly to localisation. FiLM [5] modulates channels via side input; we apply it at every decoder scale, with bottleneck cross-attention.

3. Method

Task. Given image $I \in \mathbb{R}^{H \times W \times 3}$ and prompt p , predict a grasp rectangle $\{x, y, w, h, \theta\}$ with $\theta \in [-\pi/2, \pi/2)$ [1].

Architecture (Fig. 1). GraspCLIP-D uses frozen DINOv2 and CLIP text encoders with a trainable U-Net. DINOv2 extracts tokens from four layers, reshaped to 16×16 . CLIP maps p to a 512-D embedding, projected to 768-D.

Multi-scale FiLM. At each stage i , we predict per-channel modulation from text embedding e :

$$\gamma_i, \beta_i = \text{MLP}_i(e), \quad \tilde{F}_i = \gamma_i \odot F_i + \beta_i. \quad (1)$$

MLP_i is zero-initialised so FiLM is initially identity — the decoder acts as a vision-only U-Net early on, gradually learning the language signal for stable training.

Bottleneck cross-attention. At the deepest level (16×16 tokens), visual queries attend to the sentence-level text embedding (single key/value) via 8 heads. This adds a

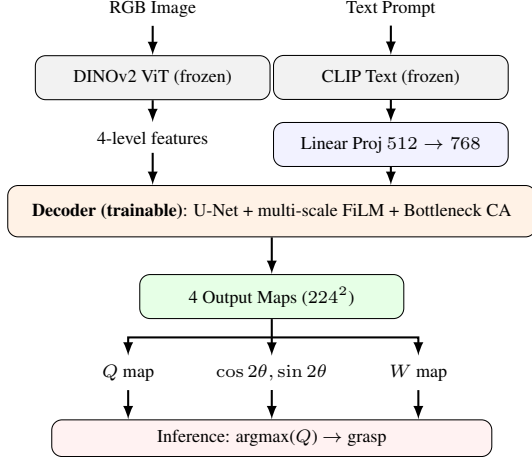


Figure 1. **GraspCLIP-D architecture.** A frozen DINOv2 ViT-B/14 produces a 4-level feature pyramid from the RGB image. A frozen CLIP text encoder embeds the prompt, projected to 768-D. The U-Net decoder applies multi-scale FiLM modulation at every stage and a single cross-attention block at the bottleneck. The head outputs four dense maps $\{Q, \cos 2\theta, \sin 2\theta, W\}$ at the input resolution, from which the grasp is decoded by arg-max on Q .

global mixing operator at negligible cost, complementing FiLM.

Output and loss. Following GG-CNN [2], the head outputs $\{Q, \cos 2\theta, \sin 2\theta, W\}$; using 2θ removes antipodal π -symmetry. We mask angle/width by GT quality:

$$\mathcal{L} = \lambda_q \mathcal{L}_Q + \lambda_a \mathcal{L}_\theta + \lambda_w \mathcal{L}_W, \quad (2)$$

with $(\lambda_q, \lambda_a, \lambda_w) = (1, 1, 0.5)$. At inference, we apply 7×7 NMS to Q , take the arg-max, and decode the rectangle.

4. Experiments

Dataset and protocol. We train on Grasp-Anything++ [8] (~ 1 M triplets, 90/10 train/val). Following [7], the top 70% most frequent prompt object labels form **Base** (seen) and the rest **New** (unseen). We report Base, New, and their harmonic mean H . A predicted grasp is correct if its IoU exceeds 0.25 with a ground-truth grasp of the referred object/part and its angular error is below 30° (mod $\pi/2$) [1].

Implementation. AdamW (lr 10^{-4} , wd 0.05), batch 64, bf16, $\pm 15^\circ$ rotation. The 10.3M trainable parameters train in ~ 4 A100 GPU-hours; the DINOv2 and CLIP encoders remain frozen.

Main results. Table 1 reports the base-to-new results. GR-ConvNet is a vision-only upper-bound, evaluated against any scene grasp without instruction grounding; it is a reference, not a directly comparable competitor. GraspCLIP-D achieves the highest H (0.60), surpassing LGD (0.47) and LLGD (0.46); our Base accuracy (0.74) notably exceeds all diffusion/consistency baselines (~ 0.52).

Ablation. Switching CLIP to DINOv2 (Table 2, rows 2 $\rightarrow$ 3) yields **+0.04** in H , confirming dense features matter

Method	Base $\uparrow$	New $\uparrow$	H $\uparrow$
GR-ConvNet [1] (vision-only) [†]	0.74	0.61	0.67
CLIP-Fusion [8]	0.40	0.29	0.33
LGD [8] (1000-step diffusion)	0.52	0.42	0.47
LLGD [3] (consistency)	0.53	0.39	0.46
GraspCLIP-D (ours)	0.74	0.51	0.60

Table 1. Main results. [†]Vision-only upper-bound, evaluated against any scene grasp; reported in [7]. LGD/LLGD use the seen/unseen split [3, 8], approximately equivalent to our base-to-new split.

Backbone*	Fusion	CA	Base	New	H
CLIP ViT-B	Concat	–	0.60	0.41	0.49
CLIP ViT-B	FiLM (last stage)	–	0.64	0.44	0.53
DINOv2 ViT-B	FiLM (last stage)	–	0.71	0.46	0.56
DINOv2 ViT-B	FiLM (multi-scale)	–	0.71	0.45	0.55
DINOv2 ViT-B	FiLM (multi-scale)	✓	0.74	0.51	0.60

Table 2. Ablation. CA: bottleneck cross-attention. *CLIP rows use ViT-B/32 and DINOv2 rows use ViT-B/14 (matched ViT-B capacity, different patch size); a fully patch-matched comparison is left to future work.

for localisation. Adding bottleneck cross-attention to multi-scale FiLM (rows 4 $\rightarrow$ 5) gives **+0.05**. The dip in row 4 vs 3 suggests distributing FiLM is beneficial only with a global mixer; row 5 confirms they are complementary.

Limitations. Our evaluation is limited to the synthetic Grasp-Anything++ benchmark; validation on real-image datasets (e.g., OCID-VLG) and physical robots remains future work. We address only planar grasps from RGB, without depth or 6-DoF extension. Finally, a controlled comparison against recent discriminative baselines under a shared protocol would further contextualise our gains.

5. Conclusion

We present **GraspCLIP-D**, pairing DINOv2 dense features with multi-scale FiLM and a bottleneck cross-attention over CLIP text. With only 10.3M trainable parameters, it offers a simple, fast alternative to diffusion-based grasp detection. Dense features and cross-attention are the key ingredients.

References

- [1] Sulabh Kumra et al. Antipodal robotic grasping using generative residual convolutional neural network, 2021. 1, 2
- [2] Douglas Morrison et al. Closing the loop for robotic grasping: A real-time, generative grasp synthesis approach, 2018. 1, 2
- [3] Nghia Nguyen et al. Lightweight language-driven grasp detection using conditional consistency model, 2024. 1, 2
- [4] Maxime Oquab et al. DINOv2: Learning robust visual features without supervision, 2024. 1
- [5] Ethan Perez et al. Film: Visual reasoning with a general conditioning layer, 2017. 1
- [6] Alec Radford et al. Learning transferable visual models from natural language supervision, 2021. 1
- [7] An Dinh Vuong et al. Grasp-anything: Large-scale grasp dataset from foundation models, 2023. 1, 2
- [8] An Dinh Vuong et al. Language-driven grasp detection, 2024. 1, 2
- [9] Chong Zhou et al. Extract free dense labels from clip, 2022. 1